

Final Exam Practice Packet

GOVT 10: Quantitative Political Analysis

Instructions: Fifty-eight questions matching the final's mix of question types. Work them closed-book at about two minutes each, then check the solutions, which are posted separately. Mark every question you miss **or guess on**, and review those topics before Saturday.

As on the exam, the only arithmetic here is predicting a value from a regression table and one difference-in-differences estimate.

1. A researcher observes that counties with more food banks have higher rates of food insecurity, and that the food banks opened before the insecurity was measured. She concludes that food banks worsen food insecurity. Which requirement for causality is she failing to satisfy?

- a) Association between the variables
- b) Temporal ordering of cause and effect
- c) A sufficiently large sample
- d) The absence of confounding

2. A study finds that legislators who receive more lobbying visits vote more often with the lobby's position. Which alternative explanation is the clearest case of reverse causation?

- a) Wealthier districts produce both more lobbying and more conservative legislators
- b) Vote records are measured with error
- c) The sample of legislators is too small
- d) Lobbyists target legislators who already agree with them

3. What does random assignment accomplish that statistical control cannot?

- a) It balances the treatment groups on variables you never measured
- b) It eliminates measurement error in the outcome
- c) It guarantees a larger sample size
- d) It removes the need to report confidence intervals

4. Mark each statement as **TRUE** or **FALSE**:

- a) Controlling for a variable measured after treatment can bias a causal estimate
- b) In a well-designed randomized experiment, you never *need* control variables for unbiasedness
- c) Natural experiments rely on treatment being as-if random
- d) An observational study can never provide any credible causal evidence

5. (Fill in the blank) Adding a control variable to a regression isolates the _____ effect of the main predictor. A control that was measured *after* treatment introduces _____ bias.

6. A state assigns a new civics curriculum to districts whose ID numbers end in an even digit. A researcher treats this as a natural experiment. What makes that defensible?

- a) The assignment rule is unrelated to anything that independently affects civics outcomes
- b) The districts volunteered, so they were motivated to comply
- c) Even-numbered districts are larger on average
- d) The researcher controls for district size

7. A canvassing experiment treats individual households. Treated households discuss the canvass with untreated neighbors, who then also turn out. What is this, and what does it do to the estimate?

- a) Attrition, which shrinks the sample
- b) Spillover, which biases the estimated effect toward zero
- c) Measurement error, which adds noise but no bias
- d) Non-compliance, which inflates the estimated effect

8. A city randomly assigns households to receive a door-knock encouraging turnout. What does the randomized control group provide that a comparison of voters and non-voters in a survey does not?

- a) A credible estimate of what the treated households would have done without the door-knock
- b) A larger sample size
- c) A guarantee that the result holds in other cities
- d) Proof that the door-knock has no unintended side effects

9. A get-out-the-vote experiment run among registered voters in one college town finds a 9-point effect. What is the main limitation on generalizing this result?

- a) Internal validity, because the randomization may have failed
- b) Reverse causation, because turnout may have caused the treatment

- c) External validity, because the sample is unrepresentative of other populations
- d) Attrition, because some participants dropped out

10. A researcher runs a two-wave study and finds that dropout between waves was far heavier among control-group participants than among treated ones. Why does this matter beyond the loss of sample size?

- a) It is not a problem; both groups still have respondents
- b) The standard errors will be too small
- c) It creates a spillover effect between groups
- d) The two groups are no longer comparable, so the remaining sample is no longer balanced by randomization

11. Consider the following model.

```
lm(donations ~ age + income + contact_hours * prior_donor)
```

	Estimate	Std.Error	t-value	p-value
(Intercept)	22.4	3.1	7.23	<0.001
age	0.45	0.09	5.00	<0.001
income	0.12	0.03	4.00	<0.001
contact_hours	3.2	0.8	4.00	<0.001
prior_donor	18.6	2.7	6.89	<0.001
contact_hours:prior_donor	-1.4	0.6	-2.33	0.020

N = 600, R² = 0.29, RMSE = 24.6

Variables: donations in dollars given this cycle · age in years · **income in thousands of dollars** · contact_hours = hours of volunteer contact · prior_donor: 0 = has not donated before, 1 = has donated before

Predict the donation for a 50-year-old **first-time** prospect with 4 contact hours and an income of \$70,000. Show your work.

12. Consider the following model.

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N = 600, $R^2 = 0.29$, RMSE = 24.6

Variables: donations in dollars given this cycle · age in years · **income in thousands of dollars** · contact_hours = hours of volunteer contact · prior_donor: 0 = has not donated before, 1 = has donated before

Predict the donation for a 50-year-old **prior donor** with 4 contact hours and an income of \$70,000. Show your work.

13. Consider the following model.

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N = 600, $R^2 = 0.29$, RMSE = 24.6

Variables: donations in dollars given this cycle · age in years · **income in thousands of dollars** · contact_hours = hours of volunteer contact · prior_donor: 0 = has not donated before, 1 = has donated before

What does the coefficient on **contact_hours:prior_donor** tell you?

- a) Prior donors give \$1.40 less than first-time prospects
- b) Contact hours and prior donation status are negatively correlated
- c) Volunteer contact reduces donations
- d) Each additional hour of contact is worth \$1.40 less among prior donors than among first-time prospects

14. In a regression table, the **Std.Error** column beside a coefficient reports:

- a) The typical size of a prediction error
- b) The share of the variance the model explains

- c) How precisely that coefficient has been estimated
 - d) The difference between the largest and smallest residual
-

15. In `lm(vote_share ~ spending + incumbent)`, the intercept represents:

- a) The predicted vote share for a challenger who spent nothing
- b) The average vote share in the sample
- c) The predicted vote share for the average candidate
- d) The share of variance the model explains

16. R-squared measures:

- a) The average size of a prediction error
- b) The share of the variance in the outcome that the model explains
- c) The probability that the model is correct
- d) The number of predictors in the model

17. A model predicting turnout reports $RMSE = 8.6$. This means:

- a) The model explains 8.6% of the variance in turnout
- b) The average residual is 8.6
- c) The coefficient on the main predictor is 8.6 standard errors from zero
- d) A typical prediction misses the observed turnout by about 8.6 percentage points

18. A residual is:

- a) An observation's outcome minus the value the model predicts for it
- b) The predicted value of the outcome
- c) The portion of the outcome explained by the model
- d) The standard error of a coefficient

19. Ordinary least squares chooses the line that minimizes:

- a) The sum of the residuals
- b) The sum of the squared residuals
- c) The largest single residual
- d) The number of residuals above zero

20. Mark each statement as **TRUE** or **FALSE**:

- a) A regression coefficient can be negative
- b) R-squared can take values greater than 1

- c) A predictor can be statistically significant even when the model's R-squared is low
- d) Two different models fit to the same data must have the same R-squared

21. A simple regression of income on years of schooling gives a coefficient of 3.1. Adding parental income drops it to 1.8. The 3.1 is best described as:

- a) The wrong answer, since the model was misspecified
- b) The causal effect of schooling
- c) The total association between schooling and income, including what runs through family background
- d) The effect of schooling among people with zero parental income

22. You omit a confounder that is positively related to both the treatment and the outcome. The coefficient on your treatment will be:

- a) Biased downward
- b) Unbiased, but less precise
- c) Biased upward, away from zero in the positive direction
- d) Unaffected

23. A randomized experiment tests whether a free school-meal program raises test scores. The researcher wants to add one control variable to tighten the estimate. Which is the only defensible choice?

- a) The number of school days the student attended during the program year
- b) The student's test score from the year before the program began
- c) Whether the student ate lunch at school during the program year
- d) The student's grade point average at the end of the program year

24. Which statement about interactions is correct?

- a) Two continuous variables can be interacted, and the coefficient describes how one variable's slope changes as the other rises
- b) Interactions are only possible between two binary variables
- c) An interaction term must always be statistically significant to be included
- d) An interaction is another name for a control variable

25. Consider the following model.

```
lm(trust ~ education + city_medium + city_large)
```

	Estimate	Std.Error	t-value	p-value
(Intercept)	44.8	2.30	19.48	<0.001
education	0.9	0.22	4.09	<0.001
city_medium	-3.6	1.60	-2.25	0.025
city_large	-8.1	1.80	-4.50	<0.001

N = 900, $R^2 = 0.11$ · Small towns and rural areas are the reference category

trust: index from 0 to 100 · education: years

Predict the trust score for a respondent in a **large city** with 12 years of education. Show your work.

26. Consider the following model.

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lm(trust ~ education + city_medium + city_large)
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N = 900, $R^2 = 0.11$ · Small towns and rural areas are the reference category

trust: index from 0 to 100 · education: years

Predict the trust score for a respondent in a **small town** with 12 years of education. Show your work.

27. Consider the following model.

```
lm(trust ~ education + city_medium + city_large)
```

	Estimate	Std.Error	t-value	p-value
(Intercept)	44.8	2.30	19.48	<0.001
education	0.9	0.22	4.09	<0.001
city_medium	-3.6	1.60	-2.25	0.025
city_large	-8.1	1.80	-4.50	<0.001

N = 900, $R^2 = 0.11$ · Small towns and rural areas are the reference category

trust: index from 0 to 100 · education: years

How do large-city residents compare to medium-city residents at the same level of education?

- a) Large-city residents score 8.1 points lower
- b) Large-city residents score 11.7 points lower
- c) The table does not permit this comparison
- d) Large-city residents score 4.5 points lower

28. A categorical variable with six levels is entered into a regression. How many dummy variables will appear in the output?

- a) Five
- b) Six
- c) Seven
- d) One

29. Consider the following model.

```
lm(trust ~ education + city_medium + city_large)
```

	Estimate	Std.Error	t-value	p-value
(Intercept)	44.8	2.30	19.48	<0.001
education	0.9	0.22	4.09	<0.001
city_medium	-3.6	1.60	-2.25	0.025
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N = 900, $R^2 = 0.11$ · Small towns and rural areas are the reference category

trust: index from 0 to 100 · education: years

If the model is re-run with large cities as the reference category instead of small towns, what happens?

- a) Nothing changes, because the model is the same
- b) The intercept and all dummy coefficients change, but the implied differences between groups are identical
- c) The R-squared changes
- d) The education coefficient changes

30. Some states adopted online voter registration between 2017 and 2021; comparable states did not. The table gives the share of eligible citizens registered. Calculate the difference-in-differences estimate.

	2017	2021
Adopted online reg.	58.3	67.9
Did not adopt	61.0	65.4

31. Some states adopted online voter registration between 2017 and 2021; comparable states did not. The table gives the share of eligible citizens registered.

	2017	2021
Adopted online reg.	58.3	67.9
Did not adopt	61.0	65.4

The difference-in-differences estimate from this table is causal only if:

- a) The two groups had the same registration rate in 2017
- b) The states were randomly assigned to adopt
- c) Absent the policy, the two groups' registration rates would have changed by the same amount
- d) The sample of states is large

32. A sample of 900 has a standard deviation of 18 and a standard error of the mean of 0.6. Which statement explains the gap?

- a) One of the two numbers must be a computational error
- b) The standard deviation describes how much individuals differ; the standard error describes how precisely the mean is pinned down, and it shrinks with sample size
- c) The standard error is always smaller because it uses a different sample
- d) The two measure the same thing on different scales

33. A t-statistic of 3.4 tells you that:

- a) The effect is 3.4 units large
- b) There is a 3.4% chance the null is true
- c) The model explains 3.4% of the variance
- d) The estimate sits 3.4 standard errors away from zero

34. A p-value of 0.001 means:

- a) There is a 0.1% chance the null hypothesis is true
- b) If the null hypothesis were true, data at least this extreme would arise about one time in a thousand
- c) The effect is very large
- d) The result will replicate 99.9% of the time

35. You conclude that a program has no effect when in fact it does. This is:

- a) A Type I error
- b) A Type II error
- c) Selection bias
- d) A confounded comparison

36. Which is a correct statement about a 95% confidence interval of [1.4, 3.8]?

- a) The procedure that produced this interval captures the true parameter in 95% of repeated samples
- b) There is a 95% probability the true parameter lies between 1.4 and 3.8
- c) 95% of observations lie between 1.4 and 3.8
- d) The parameter has a 5% chance of being exactly 2.6

37. Which of these is **NOT** something a 95% confidence interval of [-1.2, 4.6] supports?

- a) The true effect is zero
- b) The coefficient is not statistically significant at the 0.05 level
- c) The data are consistent with a true effect of 3.0
- d) The data are consistent with a true effect of zero

38. A coefficient has $p = 0.21$. What do you know about its 95% confidence interval without computing anything?

- a) It excludes zero
- b) It is narrow
- c) It contains zero
- d) Nothing can be inferred about it

39. Compared with a 95% interval on the same data, a 90% interval is:

- a) Wider, because it is less certain
- b) The same width
- c) Narrower, because it demands less confidence
- d) Undefined without knowing the sample size

40. Two studies estimate the same quantity. The one with a much larger sample will typically have:

- a) A larger coefficient
- b) A narrower confidence interval
- c) A larger p-value
- d) A higher R-squared

41. A randomized study of a job-training program reports a coefficient of 2.8, a 95% confidence interval of $[-0.4, 6.0]$, and $p = 0.09$. Mark each statement as **TRUE** or **FALSE**:

- a) The study establishes that the training has no effect on earnings
- b) The estimated effect is 2.8, and the data cannot rule out zero
- c) Because $p = 0.09$ exceeds 0.05, the estimate of 2.8 should not be reported at all
- d) Reporting only “not significant” discards the fact that effects as large as 6 remain consistent with the data

42. Mark each statement as **TRUE** or **FALSE**:

- a) A 95% confidence interval that excludes zero corresponds to $p < 0.05$
- b) Failing to reject the null hypothesis is the same as proving it
- c) With a large enough sample, a trivially small effect can be statistically significant
- d) The width of a confidence interval depends on the standard error

43. A researcher tests 40 hypotheses at the 0.05 level and reports the three that came out significant. What is the problem?

- a) Nothing; three significant results is a strong finding
- b) The tests should have used a larger sample
- c) With 40 tests, roughly two significant results are expected by chance alone, so the three cannot be interpreted at face value
- d) The researcher should have used a one-sided test

44. A study of 500,000 records finds that a policy changed turnout by 0.03 percentage points, $p < 0.001$. The best characterization is:

- a) A large and important effect
- b) A statistically significant but substantively negligible effect
- c) A non-significant effect
- d) Evidence that the policy failed

45. Group A’s mean has a 95% CI of $[42.0, 48.0]$ and Group B’s has $[47.0, 53.0]$. What can you conclude about the difference between the groups?

- a) The difference is definitely not significant, because the intervals overlap
- b) The difference is definitely significant
- c) The two groups have the same mean
- d) Overlapping intervals are not decisive; you must test the difference itself

46. A researcher has turnout for treated and control precincts and fits `lm(turnout ~ treated)`, where `treated` is 0 for control precincts and 1 for treated ones. Mark each statement as **TRUE** or **FALSE**:

- a) A two-sample t-test comparing the groups returns the same p-value as the regression's test on `treated`
- b) The regression's intercept equals the difference between the two group means
- c) Regression extends to more than two groups and to additional controls; the two-sample t-test does not
- d) Regression cannot be used here, because it requires a continuous predictor

47. The p-value beside a coefficient in `lm()` output tests whether:

- a) The whole model fits better than nothing
- b) That coefficient differs from zero in the population
- c) The residuals are normal
- d) The predictor is correlated with the other predictors

48. What does `filter(voters, age >= 25, registered == TRUE)` return?

- a) Rows where age is at least 25 **and** the voter is registered
- b) Rows where age is at least 25 **or** the voter is registered
- c) The columns `age` and `registered`
- d) One row per value of `age`

49. You want one row per state showing that state's mean income, and nothing else. Which is correct?

- a) `group_by(state) %>% mutate(m = mean(income))`
- b) `filter(state) %>% summarise(m = mean(income))`
- c) `group_by(state) %>% summarise(m = mean(income))`
- d) `select(state, income) %>% mutate(m = mean(income))`

50. Which function assigns each row to one of several categories, checking conditions in order and taking the first one that matches?

- a) `filter()`
- b) `arrange()`
- c) `n_distinct()`
- d) `case_when()`

51. `sd(income, na.rm = TRUE)` on a column with 80 missing values will:

- a) Return NA
- b) Treat the missing values as zero
- c) Replace the missing values with the mean before computing
- d) Compute the standard deviation over the non-missing values only

52. Write complete R code that keeps only respondents aged 18 or older in the data frame `survey`, then reports the mean and standard deviation of `income` separately for each level of `party`.

53. Consider the pipeline below, starting from 5,000 rows across 12 states:

```
data %>% group_by(state) %>% summarise(n = n()) %>% filter(n > 300)
```

How many rows can the result have, at most?

- a) 12
- b) 5,000
- c) 300
- d) 1

54. A national survey regresses policy support on hours of local news consumption per week. The coefficient is 1.7 with a 95% confidence interval of [0.6, 2.8]. Four researchers write the result up differently. Match each write-up to the one thing wrong with it. One flaw is not used.

Write-ups

- a) “Local news consumption causes a 1.7-point increase in policy support (95% CI: 0.6 to 2.8).”
- b) “Local news is significantly related to policy support ($p < 0.05$).”
- c) “There is a 95% probability that the true effect lies between 0.6 and 2.8.”
- d) “Local news has a large effect on policy support.”

Flaws

1. States a probability about the parameter rather than about the procedure
2. Uses a causal verb the design does not support

3. Reports significance but neither the size of the effect nor its uncertainty
4. Asserts importance without reporting the estimate at all
5. Reports the estimate but omits its direction

Answers: a = _____ b = _____ c = _____ d = _____

55. As the sample size grows large, the t-distribution:

- a) Becomes indistinguishable from the normal distribution
- b) Develops heavier tails
- c) Stops being symmetric
- d) Requires a larger critical value for the same confidence level

56. In `lm(salary ~ experience * manager)`, where `manager` is 0/1, the coefficient on `manager` is 12.0. This is the predicted salary gap between managers and non-managers:

- a) At every level of experience
- b) Averaged across all levels of experience
- c) Among employees with zero years of experience
- d) Among employees with the average level of experience

57. You add a predictor to a regression and R-squared moves from 0.20 to 0.20. What does this tell you?

- a) An error occurred; R-squared must rise when a predictor is added
- b) The new predictor explains essentially nothing beyond the existing predictors, since R-squared can never fall when a predictor is added
- c) The new predictor is collinear with the outcome
- d) The model now has too many predictors

58. A table reports, for one coefficient: estimate = 4.0, $p = 0.03$, 95% CI = [-1.1, 9.1]. What should you conclude?

- a) The effect is significant, because $p < 0.05$
- b) The effect is not significant, because the interval contains zero
- c) At least one number is misreported: $p < 0.05$ and a 95% interval containing zero cannot both be true
- d) The sample size must be too small